Project Development Phase
Scalability & Future Plan

Date	23 October 2026
Team ID	0157CY241012
Project Name	EV Market Analysis
Maximum Marks	

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Static CSV Data Ingestion: Flat CSV files require manual extract refresh when newer EV records are added.	Delays in real-time telemetry analytics and live OEM market pricing updates.	High
2	Tableau Public Concurrency Bottleneck: High concurrent user queries on complex geospatial maps may introduce latency.	Temporary render slowdowns during peak public viewing sessions.	Medium
3	Fixed Dataset Scope: Current architecture is optimized for 1,200 EV models without streaming IoT charging data.	Limits live charging grid telemetry and vehicle battery health monitoring.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Public web dashboard on GitHub Pages & Tableau Public (up to 500 concurrent views).	Migrate to dedicated Tableau Cloud / Server with auto-scaling instances and CDN load balancing to support 25,000+ simultaneous enterprise users.

2	Data Storage	Local CSV files and Tableau Hyper extract (1,200 vehicle records, ~1 MB).	Deploy cloud data warehouse (Snowflake / AWS Redshift / PostgreSQL) with partitioned data lakes capable of storing billions of real-time EV telemetry logs.
3	Performance	In-memory extract queries; sub-second response on charts, 2s on geospatial maps.	Implement pre-computed materialized aggregate views, indexed SQL queries, and distributed Redis caching for instantaneous sub-100ms response times.
4	Security	Public repository and read-only dashboard access.	Implement enterprise Role-Based Access Control (RBAC), OAuth 2.0 / SAML SSO, and Row-Level Security (RLS) to restrict sensitive OEM market intelligence.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Automated ETL Pipelines with Apache Airflow & Python to scrape daily global EV registration datasets.	Q1 2027	Eliminates manual file updates; guarantees 100% automated, continuous market intelligence.
Phase 2	Predictive Machine Learning models for EV range degradation and battery replacement cost forecasting.	Q2 2027	Empowers automotive executives and investors with 5-year battery chemistry and market demand projections.
Phase 3	IoT & Telematics Integration with live EV charging station networks (OCPP standard).	Q3 2027	Enables real-time fleet route optimization, charger availability tracking, and grid load balancing.
Phase 4	AI-Powered Conversational BI Assistant (Natural Language Querying for EV KPIs).	Q4 2027	Democratizes analytics access for non-technical policymakers and transport authorities.